



Gogineni Satya Neeraj

Roll No.:240101038

B.Tech - Computer Science and Engineering

Indian Institute of Technology Guwahati

+91-7989925889

satyaneerajg@gmail.com

n.gogineni@iitg.ac.in

Github | LinkedIn

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech.	Indian Institute of Technology Guwahati	7.92 (Current)	2024-Present
Senior Secondary	Telangana Board of Intermediate Education	97.8%	2022-2024
Secondary	Telangana Board of Secondary Education	98%	2021-2022

PROJECTS

- **CVM++ (Custom Virtual Machine + Compiler)** April 2026
Coding Club Project Github
 - Built a custom scripting language toolchain in **C++17**, implementing lexical analysis, recursive descent parsing, **AST** generation, **Bytecode** compilation, and execution through a **stack-based** virtual machine.
 - Designed a modular compiler architecture with custom bytecode, symbol table management, jump backpatching for control flow, and an interactive **CLI/REPL** supporting debugging and runtime tracing.
 - Implemented language features including variables, arithmetic expressions, conditionals, loops, and input/output handling, while validating lexer, parser, compiler, and VM execution using **CMake/CTest**.
- **MWIS Graph Optimization Solver** May 2026
Coding Club Project Github
 - Created a standalone **C++** solver for the **Maximum Weight Independent Set (MWIS)** problem on large conflict graphs, supporting datasets with up to **200,000 nodes** and millions of edges with **O(N+M)** memory complexity.
 - Applied **GWMIN**, **GWMIN2 multi-start greedy search**, **Iterated Local Search (ILS)**, and **(1,2)-swap optimization** to efficiently explore large solution spaces and generate high-quality solutions under fixed execution budgets.
 - Delivered near-optimal performance with a **0.5–2% approximation gap** on large-scale graphs, while supporting scalable input parsing, progress tracking, and configurable time-based execution.
- **Vehicle Tracking System** October 2025
Personal Project Github
 - Developed a vehicle tracking system for IITG buses and E-rickshaws using the **Django** framework with **HTML/CSS/JS** frontend, enabling real-time vehicle tracking and improved campus transportation experience.
 - Integrated features including real-time **location tracking**, bus vacancy details, route information, and E-rickshaw pre-booking using the **Geolocation API** and **Folium** for map visualization.
 - Enabled secure **registration verification** through IITG Outlook mail by **encoding user** information with **timestamp**-based validation for authenticated access.
- **Optimized RV32IM RISC-V Coprocessor** March 2026
Course Project, Mr.Lokesh Siddhu Github
 - Built an **RV32IM RISC-V processor** in **Verilog HDL** with matrix multiplication and **CORDIC-based** accelerators for neural network inference, integrating **ALU**, **MUL**, **DIV**, **MAC**, **DMA controller**, **UART communication**, and instruction/data memory modules.
 - Verified the system using **Xilinx Vivado** on the **Nexys A7 FPGA board** with **HEX-based** memory initialization, **UART output transmission**, and subsystem-level simulation/testing for reliable hardware execution.

TECHNICAL SKILLS

- **Programming:** C, C++, Python, SQL
 - **Web technologies:** HTML, CSS, JavaScript, React JS, Node JS, Django*, MySQL, MongoDB*
 - **ML Workflow:** Numpy, Pandas, Scikit-learn, Tensorflow*, Keras*, Pyspark, GGplot, Databricks
 - **Tools:** Git, Github
- * Elementary Proficiency

KEY COURSES TAKEN

- **Computer Science:** Data Structures and Algorithms, Database Management Systems, Computer Architecture and Organization, Design and Analysis of Algorithms
- **Mathematics:** Linear Algebra, Basic Calculus, Discrete Mathematics, Probability & Random Processes

ACHIEVEMENTS

- **Codeforces:** Achieved a peak rating of **1566** in competitive programming contests (**profile**)
- **Codeforces Round 1101 (Div. 2):** Secured a global rank of **1293** among **22,000+** participants
- **Codechef:** Attained a maximum rating of **1700**, earned the **3-Star Coder** badge (**profile**)
- **Codechef Starters Round 241:** Ranked **365** in Division 2 among **2,000+** participants

EXTRACURRICULAR ACTIVITIES

- **Volleyball Team – IIT Guwahati,** Represented IIT Guwahati at the national-level Inter IIT Sports Meet 2025
- **Spardha,** Secured a **Gold Medal** in the inter-hostel sports tournament 2025
- **Spirit,** Earned a **Silver Medal** in institute-level intra-college sports events 2025
- **Eternia,** Won a **Gold Medal** in the annual intra-college sports tournament 2025